Gotu AI Agriculture

Where Tech with Intelligence Meets Soil

1 Tomato

1.1 Land Preparation

- **Site selection:** Heavy soils are documented as requiring raised seedbeds to avoid waterlogging, whereas flat beds are permitted in sandy soils. Irrigation access is documented for nursery plant establishment. Suitable field topography, slope, sunlight exposure, soil texture, soil depth, and previous cropping history are not specified in the provided sources.
- **Soil requirements:** Heavy soils and sandy soils are noted in relation to nursery bed preparation methods. Documented soil pH, electrical conductivity, organic matter content, soil fertility, nutrient status, and salinity levels are not specified in the provided sources.
- **Soil testing:** Soil testing is recommended prior to sowing and land cultivation to control over-fertilization, assist in proper crop planning, achieve double economic benefits, and determine soil nutrient levels. Specific soil sampling procedures, sampling depth, sampling locations, sample preparation, laboratory testing methods, interpretation of results, and soil-test-based fertilizer recommendations for tomato (*Solanum lycopersicum*) are not specified in the provided sources.
- **Field clearing and initial preparation:** General land cultivation prior to sowing is recommended. Removal procedures for weeds, crop residues, stones, previous crop remains, and initial field cleaning operations are not specified in the provided sources.
- **Primary tillage:** General land cultivation is recommended before sowing. Specific ploughing methods, equipment, tillage depth, number of operations, timing, and documented effects on soil structure or root development are not specified in the provided sources.
- **Secondary tillage:** The surface of seedbeds must be smooth and well levelled during bed preparation. Specific secondary tillage operations, including harrowing, clod breaking, pulverization, sequence, and timing, are not specified in the provided sources.
- **Soil amendments and organic matter:** Sieved farmyard manure mixed with fine sand is added to the nursery seedbed surface. Application rates per unit area, incorporation depth, quantities of farmyard manure, compost, green manure, lime, gypsum, or other soil amendments for tomato land preparation in the main field are not specified in the provided sources.

- **Fertilizer application during land preparation:** Soil testing is advised to assess soil nutrient status and prevent over-fertilization. Specific tomato basal fertilizer sources, application rates, nutrient quantities per unit area, placement methods, incorporation procedures, and timing relative to transplanting in the main field are not specified in the provided sources.

- **Soil drainage and water management:** Raised nursery beds of 10–15 cm height or broad bed furrows are used to prevent waterlogging in heavy soils. Nursery beds are initially irrigated using a water can equipped with a sprayer before transitioning to standard irrigation water. Soil moisture is maintained by covering beds with dry straw, grass, or sugarcane leaves until germination, followed by seedling hardening through slight water withholding during the final nursery week. Main field drainage channel specifications, raised field structures, and irrigation infrastructure layout are not specified in the provided sources.

- **Field layout and land leveling:** Nursery beds require a smooth and well-levelled surface, with a 70 cm distance maintained between adjacent beds to facilitate watering, weeding, and operational access. Main field dimensions, plot layout, slope management, land leveling procedures, field boundaries, and access paths are not specified in the provided sources.

- **Bed, ridge, and furrow preparation:** Nursery beds are prepared as raised beds measuring 3 m
(×
)$0.6mwithaheightof10--15cm, spaced70cmapart, orasbroadbedfurrows.Raisedbedsarerecomm$

- **Raised beds and protected cultivation:** Land preparation practices covering open-field nursery raised beds and broad bed furrows are documented. Greenhouse, polyhouse, or other protected cultivation systems are not specified in the provided sources.

- **Soil solarization and disinfestation:** Nursery seedbeds are drenched first with water and subsequently drenched with Bavistin at a rate of 15–20 g/10 L of water to prevent seedling mortality from damping-off disease. Prior to sowing, seeds are treated with fungal culture of *Trichoderma viride* at 4 g/kg of seed or Thiram at 2 g/kg of seed. Physical soil solarization, chemical soil disinfestation, active ingredients for main field soil treatment, covering materials, duration, and temperature requirements are not specified in the provided sources.

- **Weed, pest, and disease management before planting:** Inter-bed spacing of 70 cm is provided to allow weeding operations. Damping-off disease is controlled through seedbed drenching with Bavistin (15–20 g/10 L of water) and seed treatment using *Trichoderma viride* (4 g/kg of seed) or Thiram (2 g/kg of seed). Pre-planting field sanitation, weed control products, crop residue management, soilborne pest management, and pre-plant chemical control rates for the main field are not specified in the provided sources.

- **Water and irrigation system preparation:** Irrigation facilities are documented as necessary for tomato seedling production. Seedbeds are initially watered using a

water can with a sprayer, followed by regular irrigation water application as needed. Installation of irrigation channels, drip irrigation pipelines, emitters, filtration systems, and operational checks are not specified in the provided sources.

- **Environmental and seasonal considerations:** Sowing and field preparation schedules vary by season: Kharif nursery sowing occurs in May to June (or June to July for autumn-winter crops) with transplanting in June to July; Rabi nursery sowing occurs in September to October with transplanting in October to November; spring-summer nursery sowing occurs in November (or January to February) with transplanting in February to March; and hill region sowing occurs in March to April. Dry straw, grass, or sugarcane leaf covers are placed over seedbeds to maintain required temperature and soil moisture during germination. Specific rainfall thresholds, soil moisture workability limits, and weather effects on main field tillage timing are not specified in the provided sources.
- **Timing and scheduling:** Seedbed drenching with Bavistin is conducted prior to sowing after initial watering. Straw or grass covers are removed immediately after germination is complete. Seedling hardening is scheduled during the final week in the nursery by slightly withholding water. Seedlings reach field readiness within 4 weeks of sowing when they develop 5–6 true leaves. Recommended timing for field clearing, primary ploughing, soil amendment application, fertilizer incorporation, and final main field preparation relative to transplanting is not specified in the provided sources.
- **Field readiness assessment:** Nursery seedlings are assessed as ready for field transplanting when they attain 5–6 true leaves within 4 weeks of sowing. Documented main field readiness indicators, including soil tilth, soil moisture thresholds, bed stability, weed status, and main field irrigation readiness, are not specified in the provided sources.
- **Effects on crop performance:** Soil testing prevents over-fertilization and supports optimal crop planning for double economic benefits. Seedbed drenching with Bavistin and seed treatment with *Trichoderma viride* or Thiram prevent seedling mortality caused by damping-off. Raised beds prevent waterlogging in heavy soils. Quantitative relationships between tillage depth, root penetration, soil structure, transplant establishment, and tomato fruit yield are not specified in the provided sources.
- **Conflicting or incomplete recommendations:** Documented recommendations exhibit minor variations in sowing schedules across sources: one source specifies Kharif seed sowing in May to June for transplanting in June to July, whereas another specifies June to July for the autumn-winter crop. For the spring-summer crop, seed sowing is documented in November in one source and January to February in another, with transplanting in February to March. Seed requirement is specified as 250–300 g/ha of land. Both sources recommend raised beds or broad bed furrows for nursery preparation, but comprehensive land preparation practices, tillage depths, fertilizer doses, and amendment rates for main field establishment are incomplete and not specified in the provided sources.
- **Integrated workflow:** The documented sequence begins with soil testing to assess nutrient levels, followed by initial land cultivation. For nursery establishment,

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